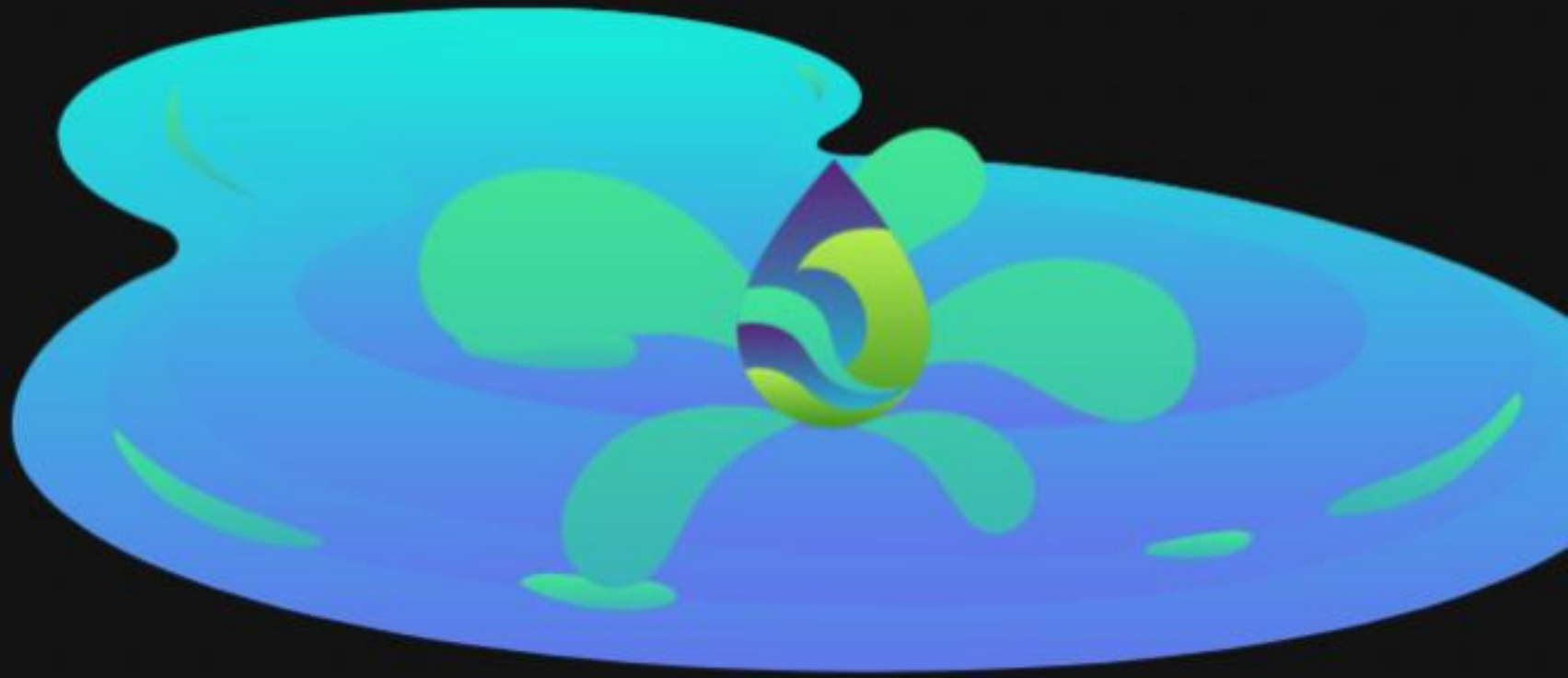


Kathryn Dunstan
Presents

Load Cell
Based
Floating
Pipe Water
Level
Detector



A²H₂FLOAT

AU² H₂ Float

Improving design and method for monitoring water levels remotely for early warning indication of flooding.

- **By Optimizing Cost**
- **Low Power Requirement**
- **Environmentally Friendly Application**

Continuous Water-Level Sensor Based on Load Cell and Floating Pipe

research done by

Sheng-Wei Wang; Chen-Chia Chen*; Chieh-Ming Wu; Chun-Ming Huang

Archimedes Principle

F_B = Weight Displaced

$$F_B = \rho_w g(V)$$

$$F_B = \rho_w g(\pi r^2 \cdot d)$$

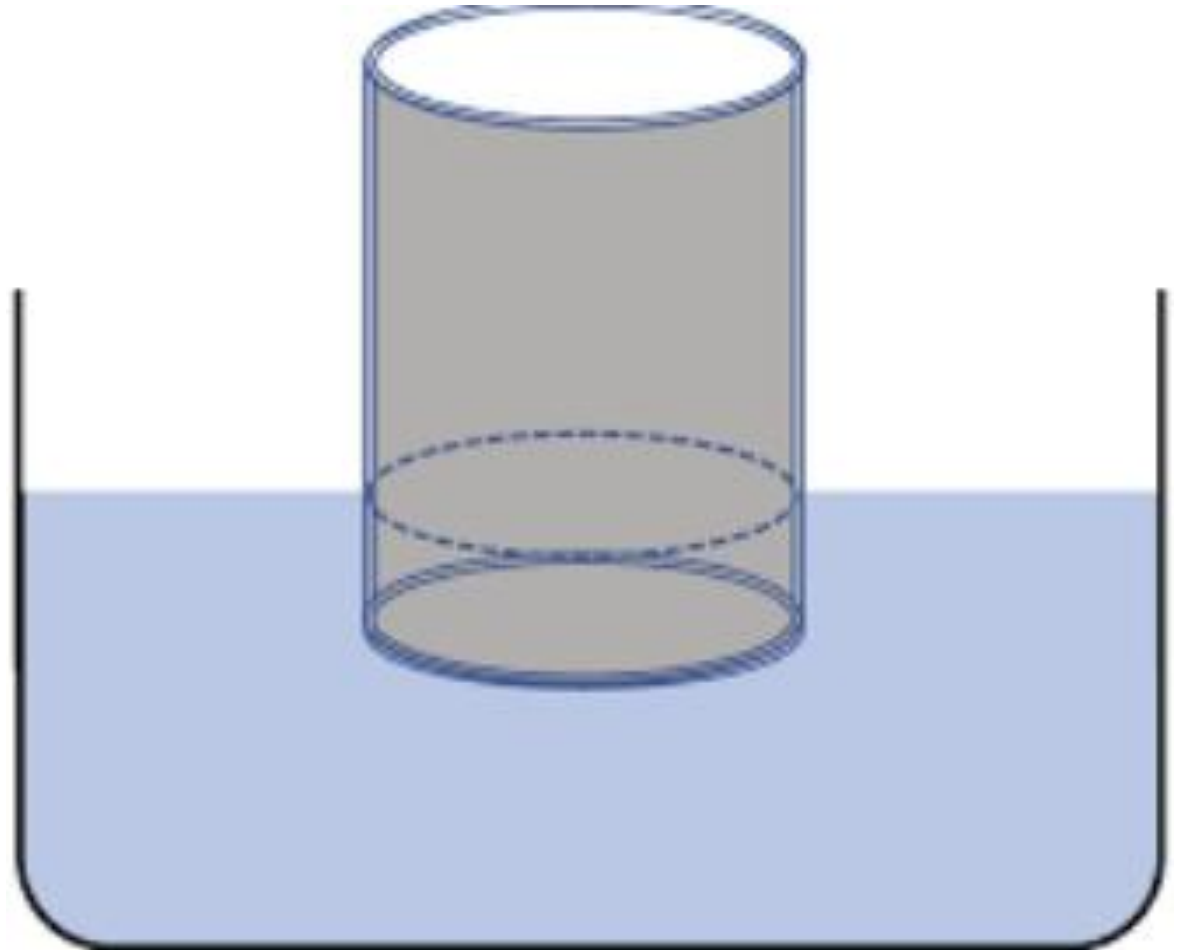
F_B = Buoyant Force

ρ = density of the water

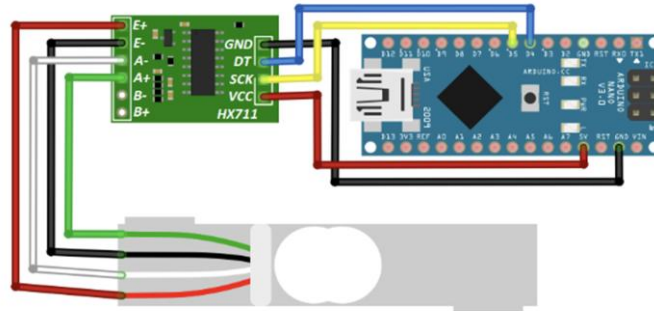
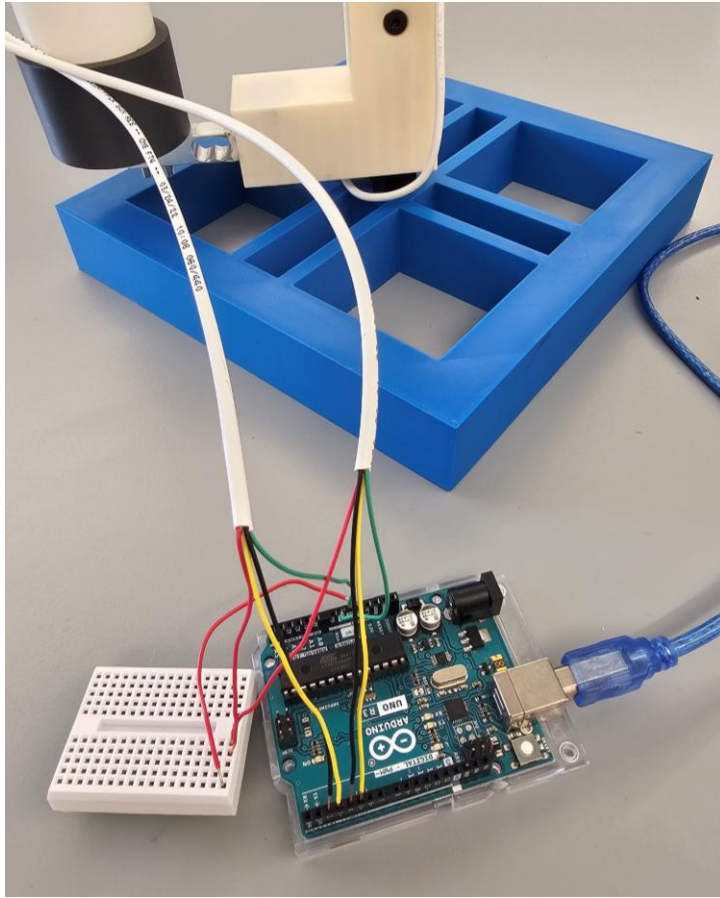
g = acceleration due to gravity

V = Volume of the displaced fluid

d = distance object is displaced



Floating Pipe and Load Cell Setup



LoRa WAN :

- Low Power, Wide Area (LPWA) networking protocol
- Wirelessly connect battery operated devices to network server via standard IP connections
- Bi-directional communication
- Converting RF packets to IP packets
- Uses **Long Range RF**

